Alaa Anani

Saarbrücken, Germany

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Research Interests

Mechanistic understanding and **certified robustness** of **vision** and **language** foundation models.

Education

Saarbrücken, Germany

Max Planck Institute for Informatics & CISPA Helmholtz Center *Aug 2024 – Aug 2028 (Expected)*

Ph.D. in Computer Science
Area: Trustworthy machine learning, with a current focus on **interpretability of LLMs**.
Supervisors: Prof. Dr. Bernt Schiele and Prof. Dr. Mario Fritz

Saarbrücken, Germany

Saarland University *Oct 2021 – May 2024*

M.Sc. in Computer Science – GPA: 1.6 (Best is 1.0)
Thesis: Adaptive Hierarchical Certification for Segmentation using Randomized Smoothing.
Supervisors: Prof. Dr. Bernt Schiele and Prof. Dr. Mario Fritz

Cairo, Egypt

The American University in Cairo *Sept 2016 – June 2021*

B.Sc. in Computer Engineering – GPA: 3.96/4.0 (Summa Cum Laude), Minor: Mathematics
Thesis I: Mixture of Experts for Human Activity Classification using Semantic Fusion.
Thesis II: On Ramsey Theory and its Computer Science Algorithmic Applications.
Supervisors: Prof. Cherif Salama, Prof. Mohamed Moustafa and Prof. Daoud Siniora

Publications

[1]: **A. Anani**, T. Lorenz, B. Schiele, M. Fritz, and J. Fischer. "Certified Circuits: Stability Guarantees for Mechanistic Circuits." In *International Conference on Machine Learning (ICML)*, 2026.

[2]: **A. Anani**, T. Lorenz, M. Fritz, and B. Schiele. "Pixel-level Certified Explanations via Randomized Smoothing." In *International Conference on Machine Learning (ICML)*, 2025.

[3]: **A. Anani**, T. Lorenz, B. Schiele, and M. Fritz. "Adaptive Hierarchical Certification for Segmentation using Randomized Smoothing." In *International Conference on Machine Learning (ICML)*, 2024.

Research Experience

SpreeAI **Los Angeles, USA**

Machine Learning Research Engineer *Oct 2021 – Dec 2023*

Topics: 3D head tracking, portrait illumination transfer, face texture transfer, and head restoration.
Reference: Dr. Minh Vo

École Polytechnique Fédérale de Lausanne – MLO Lab **Lausanne, Switzerland**

Summer@EPFL Intern *July 2021 – Oct 2021*

Topic: Interpretability of a COVID-diagnosis ViT from lung ultrasound images.
Reference: Dr. Mary-Anne Hartley

Dell Technologies – Data Office **Cairo, Egypt**

Data Science Research Intern *Dec 2020 – June 2021*

Topics: Technology forecasting and organized Egypt's first AI hackathon (*C4F2*).
Reference: Eng. Steve Todd

Synapse Analytics – Data Science Team **Cairo, Egypt**

Data Science Intern *July 2020 – Aug 2020*

Topic: Credit scoring with time-series analysis.

The American University in Cairo **Cairo, Egypt**
Undergraduate Researcher *Sept 2019 – June 2020*
Topic: Deep reinforcement learning for traffic intersection control.
Reference: Prof. Cherif Salama

Project Highlights

Practical Machine Deep Learning (**GitHub**)
Implemented CNNs from scratch in Python. *Fall 2020*

AUC Robotics Club
Designed and implemented the control system for a remotely operated underwater vehicle [*Vader I-II*]. *Spring 2020*

Computer Architecture
3-Stage pipe-lined CPU with interrupts support in Verilog. *Fall 2018*

Teaching

Saarland University & MPI for Informatics **Saarbrücken, Germany**
Graduate Teaching Assistant *Oct 2021 – March 2022*
Course: Automated Reasoning
Reference: Prof. Uwe Waldmann

The American University in Cairo **Cairo, Egypt**
Undergraduate Teaching Assistant *Feb 2021 – June 2021*
Course: Graph Theory
Reference: Prof. Daoud Siniora

The American University in Cairo **Cairo, Egypt**
Undergraduate Teaching Assistant *Feb 2021 – June 2021*
Course: Discrete Mathematics
Reference: Prof. Daoud Siniora

The American University in Cairo **Cairo, Egypt**
Undergraduate Teaching Assistant *Feb 2021 – June 2021*
Course: Embedded Systems Lab
Reference: Dr. Suzanne Safwat

Awards

2026: Gold Reviewer Award (top 25%) *International Conference on Machine Learning* (*ICML*)
2024, 2025: Best Poster Award (×2), ELSA General Assembly *European Lighthouse of Secure and Safe AI*
2021: CS@Max Planck Highly Selective Doctoral Grant *CS@Max Planck*
2021: Dr. Abdel Rahman El Sawy Award in Engineering ($1K) *AUC*
2021: 4th Place, WomenHack Data Science Hackathon *McKinsey*
2019: Top-10 ROV Team in the Middle East *MATE ROV Regional Competition*
2019: 4th Place in the Minesweepers Competition *Minesweepers*
2016–2021: Dean's List for Academic Standing *AUC*
2016: Egyptian Government PSSF Scholarship (~$80K) *Egyptian Government*

Extracurricular

2017–2020: Led the control team in the ROV competition *MATE ROV*
2018: Designed university-level robotics programs and competitions as an Academic Director *AUC Robotics*

Skills

Programming: Python, C, C++, CUDA C, ROS, Verilog, MATLAB, LabVIEW
Frameworks: PyTorch, CUDA, TensorFlow, Docker, Kubeflow, UNIX, Gazebo, SUMO, LaTeX
Languages: English (Fluent), Arabic (Fluent), German (A2)